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POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
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MarikinAlert

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I. PROJECT PROPOSAL

1. Problem Statement

Modern urban centers like Marikina City face an increasingly complex landscape of multi-hazard risks, ranging from the perennial threat of riverine flooding to the catastrophic potential of the West Valley Fault and structural fires in high-density areas. Despite being a model for disaster preparedness in the Philippines, Marikina's existing emergency protocols—such as the Rescue 161 and 911 Marikina hotlines—face a significant vulnerability during high-intensity events: their reliance on a centralized telecommunications backbone. During what is termed "Data Chaos," the simultaneous outburst of 500 to 1,000 reports can lead to network congestion or total infrastructure collapse, effectively blinding the Marikina City Disaster Risk Reduction and Management Office (MCDRRMO) from real-time situational awareness.

MarikinAlert is a hyper-localized disaster reporting and response application designed exclusively for Marikina City to mitigate these failures. The application's core innovation lies in its internet-independent infrastructure. By utilizing a distributed network of local Wi-Fi nodes, the system establishes a "city-wide intranet" that allows residents to report various emergencies—including fires, medical crises, and structural collapses (trapped victims)—even when global internet connectivity is severed.

The system moves beyond simple communication by integrating an Automated Triage Engine capable of interpreting unstructured "Taglish" inputs. This AI-driven triage automatically categorizes and prioritizes reports based on life-threatening keywords like *"sunog"* (fire) or *"naipit"* (trapped), ensuring that high-acuity needs are immediately flagged for dispatch. Furthermore, to avoid a single point of failure, MarikinAlert utilizes a Distributed Database Architecture. This ensures that any report received by a single node is mirrored across all other admin nodes in real-time, providing a resilient and synchronized "Single

Source of Truth" for emergency responders across all 16 barangays.

2. Objectives

- **Local Resilience:** To establish a communication bridge between residents and authorities via local Wi-Fi hotspots without requiring global internet access.
- **Automated Triage:** To implement an AI-driven categorizer that can interpret "Taglish" (Tagalog-English) reports to automatically assign priority and emergency categories.
- **High-Volume Reliability:** To prove the system can handle an outburst of 500 to 1,000 simultaneous reports without crashing, maintaining accurate analysis during peak disaster events.
- **Distributed Synchronization:** To ensure that any report received by one node (admin device) is reflected across all other nodes in the network, creating a "city-wide intranet" effect.

3. Target Users

A. Primary Users

- Marikina Residents - Our primary users are the city's residents who are witnessing or experiencing a disaster. The system is optimized for speed and ease of use, minimizing the cognitive load required to report an emergeny
    - Demographic: Residents of Marikina City
    - State of Mind: High Stress, Panic, Urgency
    - Technical Literacy: Varied
    - Language: Taglish

  They provide the raw message, sender name, and contact number. They act as sensors on the ground, reporting incidents like Fire, Collapse, or Medical Emergencies.

B. Secondary Users

- The Admins/System Operators - These are Incident Triage Officers or Dispatchers. They are equipped with laptops connected to local public wifi acting as a "Node" in the distributed network. Their responsibility includes verifying reports and coordinating with rescue agencies (BFP, PNP, LGU). Specified roles are:
    - To monitor the dashboard to view incoming reports
    - To prioritize incidents based on color codes (Red is critical, Yellow for high)
    - Verifies the legitimacy of the report.
    - Acts as a relay forwarding confirmed specific threats to agencies.

C. Tertiary Users

- Emergency Response Agencies - While they may not login into the dashboard directly, they are the downstream consumers of the data processed by MarikinAlert.
    - Bureau of Fire Protection (BFP) - For Fires
    - Local Medical Teams/Ambulances - For Medical
    - Engineering Department/DPWH - For infrastructure categories.

II. SYSTEM OVERVIEW

1. Features

- **Offline Reporting Node:** Operates on a local Wi-Fi intranet, allowing reports to be submitted even when the city's main internet backbone is severed.
- **Automated Taglish Triage:** An integrated AI service that understands natural local language (Tagalog-English) to filter and categorize emergencies without manual human sorting.
- **Distributed Node Mirroring:** Every station laptop acts as an independent node. When Node A receives a

report, it automatically updates Nodes B and C across the city network, ensuring data consistency and redundancy.

- **Real-Time Dispatch Dashboard:** A color-coded administrative interface that dynamically prioritizes life-threatening reports (e.g., Building Collapse) at the top of the queue.
- **IsVerified Protocol:** A workflow for dispatchers to mark legitimate reports, preventing duplicate responses and ensuring accuracy for emergency agencies.

2. Functionality

- **Ingestion:** A resident connects to the "Marikina Public Wi-Fi" and accesses the submission form.
- **Processing:** The system passes the RawMessage to the IDisasterTriageService. The service identifies keywords (e.g., *"sunog"*, *"trap"*) and maps them to a ReportCategory and ReportPriority.
- **Persistence:** The finalized DisasterReport entity is stored in the local SQL Server database via the Repository Pattern.
- **Display:** The Admin Controller fetches unverified reports and pushes them to the live dashboard using asynchronous updates.

III. TECHNOLOGY STACK

Component	Technology
Framework	**ASP.NET Core MVC** (.NET 8.0)

Language	**C#**
Data Access	**Entity Framework Core** (EF Core)
Database	**SQLite**
Frontend	**Razor Views**, **Bootstrap**, **JavaScript/AJAX**
Architecture	**Repository Pattern** & **Dependency Injection (DI)**
Testing	**NUnit** & **Moq**

IV. INSTALLATION/SETUP INSTRUCTIONS

A. Prerequisites
   a. .NET 8.0 SDK
   b. Visual Studio 2022 or VSCode
   c. Git.

B. Step-By-Step Guide
   a. Clone the repository
      i. git clone https://github.com/hopeiea/MarikinAlerttt.git
      ii. cd MarikinAlert.web
   b. Setup Database
      i. Open Terminal
      ii. Run these commands:
         1. dotnet tool install -global dotnet -ef
         2. dotnet ef database update

C. Run the Application
   a. dotnet run

D. Admin Access
    a. Login Page: /Admin/Login
    b. Username: admin
    c. Password: marikina123

V. USER GUIDE

For the Resident (Client Side)

1. **Connect**: Switch your device Wi-Fi to the "Marikina Local Public Wi-Fi" network.
2. **Access**: Navigate to the provided local URL (e.g., http://marikina-alert.local).
3. **Report**: Fill out the **Sender Name**, **Location**, and **Raw Message**.
    - *Tip*: Use plain language like *"May malaking sunog dito sa tapat ng Mercury Drug"* or *"Naipit po kami sa guho."*
4. **Submit**: Click the "Submit Emergency Report" button. You will receive a confirmation that your report is being triaged.

For the Dispatcher (Admin Side)

1. **Login**: Access the admin portal via http://marikina-alert.local/Admin.
2. **Monitor**: Watch the live dashboard. Reports will appear automatically.
    - **Red**: Critical (Immediate Life Threat)
    - **Yellow**: High Priority (Fire/Infrastructure Damage)
3. **Verify**: Click the **"Verify"** button after confirming the report with local scouts or radio.
4. **Forward**: Copy the location and category details to relay to the **BFP**, **PNP**, or **MCDRRMO** rescue teams.

VI. SCREENSHOTS/DIAGRAMS

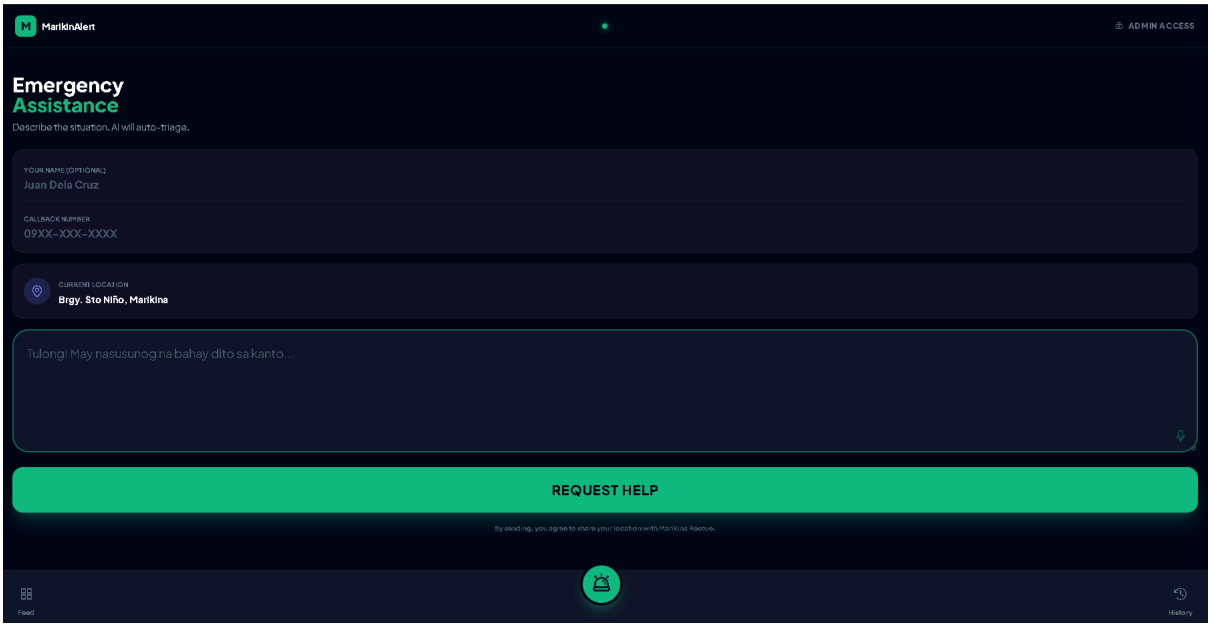


Figure 1: Landing Page: Users can immediately report an incident

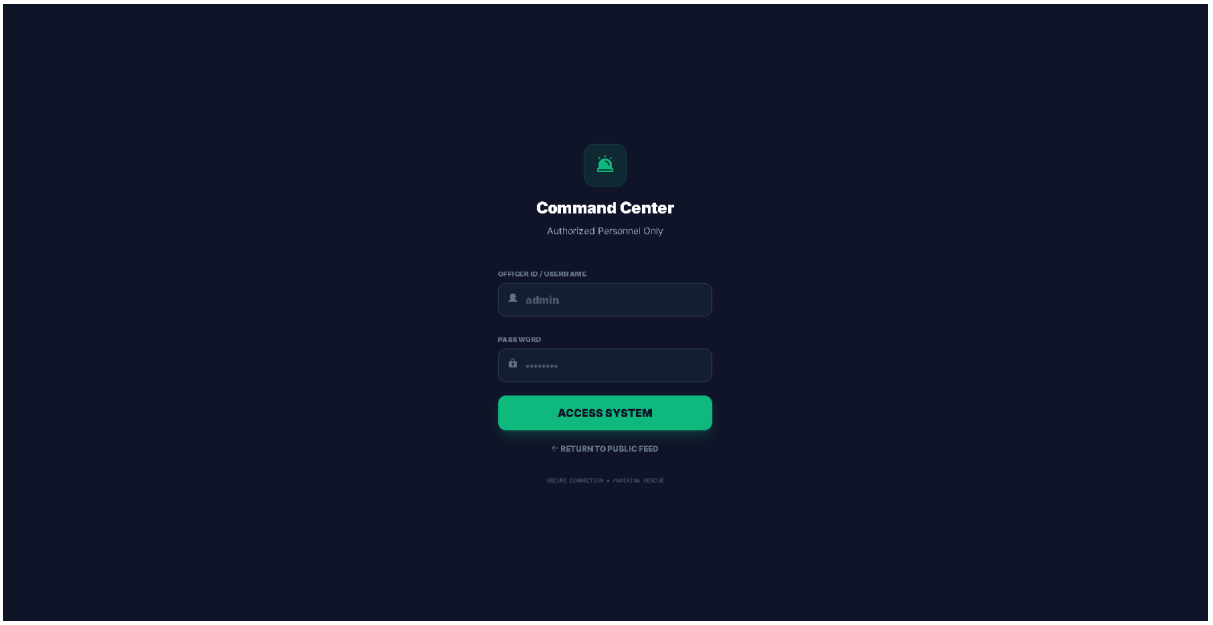


Figure 2: Admin Login Page

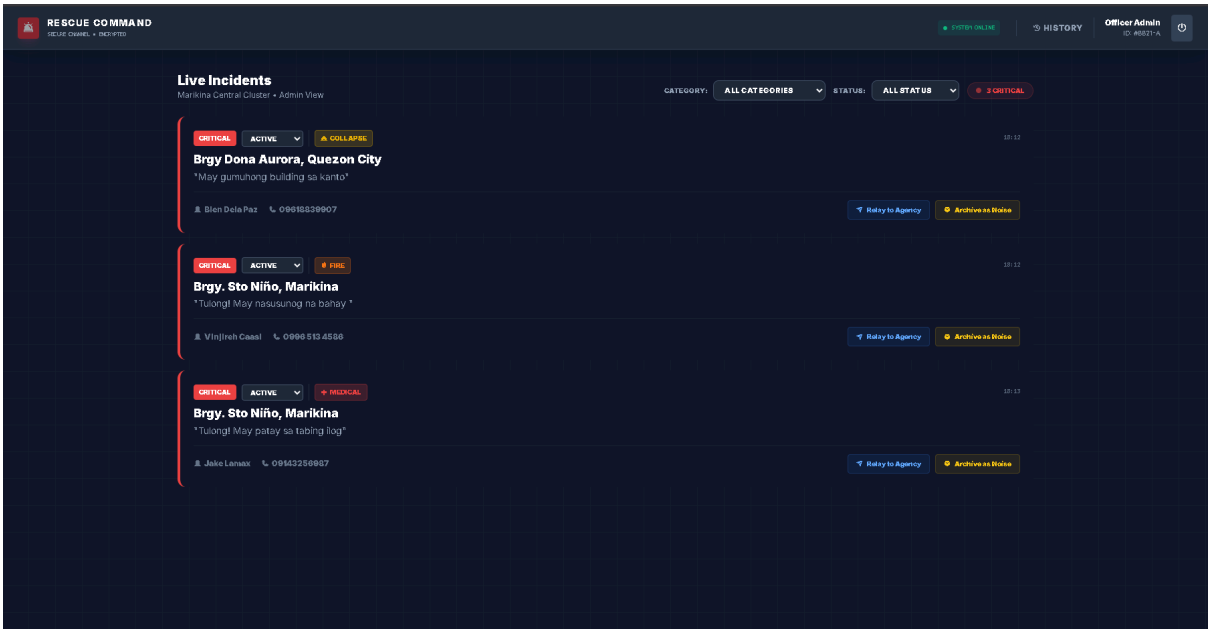


Figure 3: Admin Dashboard

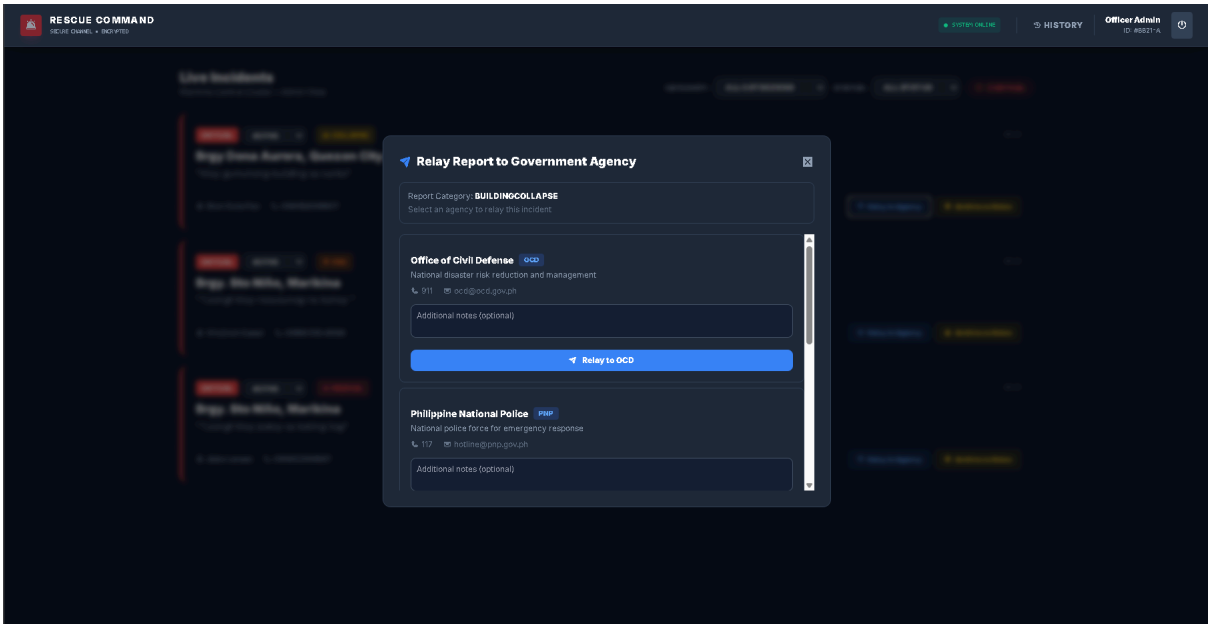


Figure 4: Relay to Agency Feature



Figure 5: UI/UX on End User Device (Mobile)

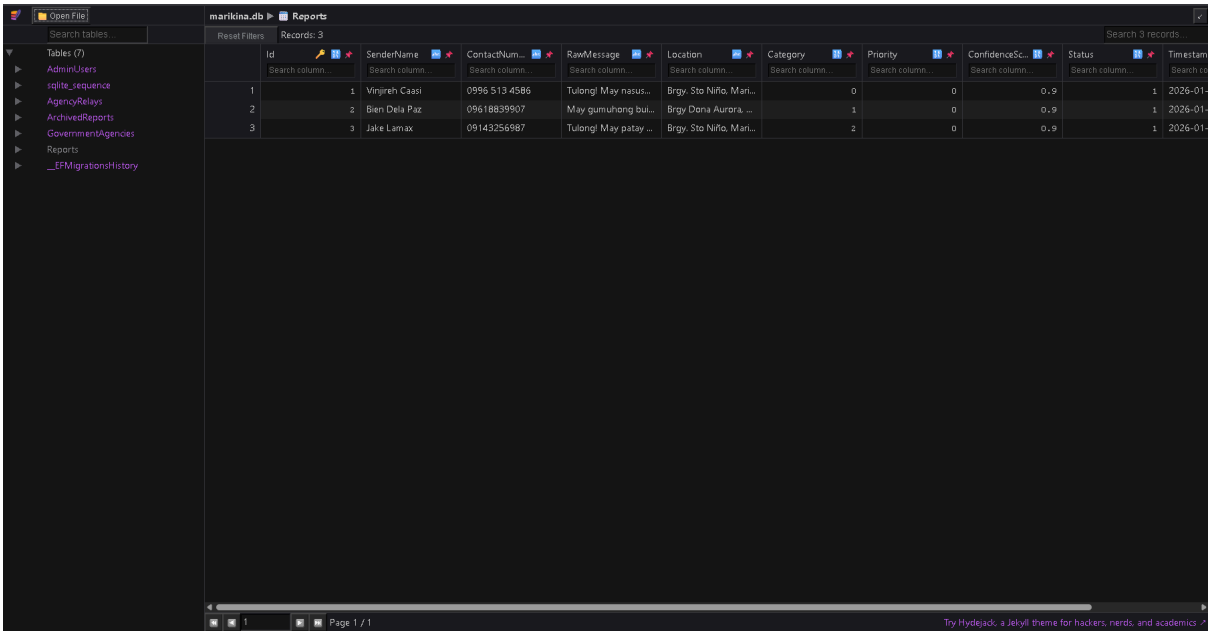


Figure 6: Database Input

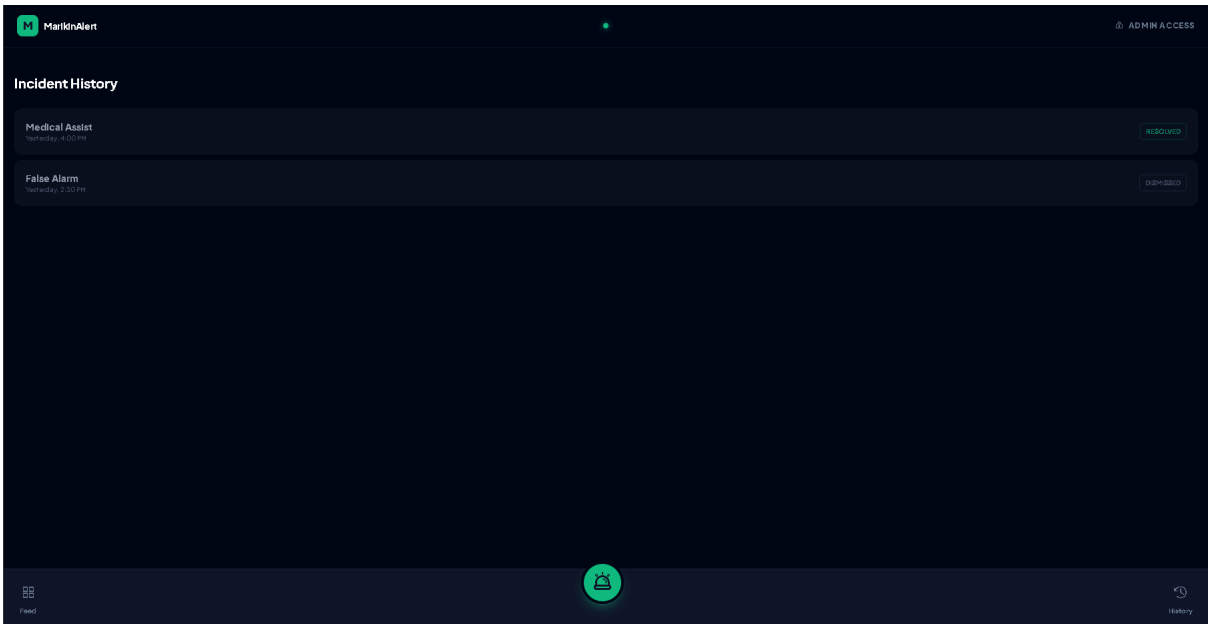


Figure 7: History of Reports/Incidents